

Probability Basics from start to distributions

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1 Introduction

Let's start with the intuition from start in which will see the problem how the topic was invented to solve it and then its drawback then next topic which solve that issues then so own.

Definition 1 (Uncertainty) *Let's can an example consider any specific goal in life like i take a example i have a question in mind "will i do PHD in life ?" this the outcome of this specific question i don't know this specific question under my observation is known as a "event" and the result which i want to get is known as "outcome"*

like this any thing in life can be consider an example of uncertainty because we don't what is the result

Now what we want is that we want to calculate that whether this event will happen or not (means what should or will be the output)

This will take us to our next topic name as Probability

Definition 2 (Probability) *What we want is that what will be the outcome of any event so we want to know what will be the chance that it will happen or not. So that measuring the chance of all possible outcomes the calculation or measurement of it will be called "Probability"*

consider it like quatification of all possible outcome if there is student if he is brilliant the chance that he will pass is more then he will fail so can say that it is "80%" chance that he will pass and "20%" of fail if he gets ill so total will be 100

Axiom of probability:

1. that it remain between 0 and 1. (1 represent it will happen while 0 it will not happen so either the event is going to occur or not atleast so it works)
 2. the total of all the possible outcome should equal to one (as it will make sense because there can be nothing beyond happening of some event)
- "Summary": if the uncertainty is Unknown value of any event; about which

we are not sure then probability help you to make a guesses of all the possible outcomes of that unknown value what it should be.

Definition 3 (Expectation) *"The average value of the outcome that occurs most of the time" it's the weighted average of all the possible outcomes, not just the most frequent one. "Weighted Average": Is just like the average value but weights attached to it means the value that occurs more have higher weight then the other that occurs less have the lower weight*

Formula of Expected Value:

$$E[x] = \sum (value * probability of value)$$

Now we should move to the next step which is sample space.

Definition 4 (Sample Space) *As we said, every event has a defined number of possible outcomes. Like in dice if we throw it can give us only 6 possible outcomes from 1 to 6 dots at a time. If we take all the possible outcomes and keep in a set, it will consider as a Sample space.*

$$Samplespace of dice = \{1, 2, 3, 4, 5, 6\}$$

Now there is a problem that not every event has a numerical outcome like coin toss, it gives head or tail, but we want numerical values The reason is that we can apply probabilities and want to work with the results which can be done with numerical values only.

So, to this problem we have next Random Variables

Definition 5 (Random Variable) *The Random Variable is a function that assigns a numerical value to each outcome of an event.*

Sample Space of coin toss: H,T

Random variable=X

$$X(H) = 0$$

$$X(T) = 1$$

Now let's talk about the reason it is called "Random Variable" because we don't want value it will be as the outcome so it is random to use and value varies that's why it is a random variable. But remember as we talk before that each outcome have its own probability(chances of that outcome to be occur) it can vary for each outcome some have more probability and some not.

You know what is fun part about a random variable it is just like other variable but from a specific set and that is the sample space of an event.

For Example:

"it is like peaking up a ball but from a specific container if there 5 balls 2 blue,1 red, 1 green, 1 yellow in that container the all 5 balls are the sample

space of it and assigning the value to each color is done by a function called random variable blue=1, green=2, yellow=3, red = 1
you can see after random variable we have 4 values because blue occur 2 times so the probability of blue $2/5$, while for other it is $1/5$ ”

From above you can see that each random variable have the probability attach to it and there sum is equal to 1 follows the axioms of Probability.

I hope so i clear the topic which i want to explain it to you.

There are two Types of ”Random Variable”:

1. Discrete Random Variable.
2. Continous random Variable.

Definition 6 (Discrete Random Variable) *It is when the outcomes are countable as we explain in the above example so i am not focusing on it.*

Formula for discrete Random Variable:

$$\sum P(xi) = 1$$

Definition 7 (Continous Random Variable) *It is when the outcomes are not countable like height of the person it can't given specific value on scale it can 5.1 and another with more precious like 5.11112 so it change with precious of the scale we can't say exact value*

The reason we use range in continous is that there infinite many numbers in a value from to numbers like 3 and 4 there is 3.12 or 3.34565 it can goes on and on and if the each value is assigned with some probability as there are infinite many value so the sum of probability will be infinite which break the axioms rule according to which it should be 1. therefore we break the variables in range and assigned probabilities to it like height of people from 0 to 1 , 1 ot 2, 3 to 4 , 5 to 6, 6 to 7 the person of height 5.8 will be kept in range 5 to 6 and each range will assigned with a probability to it which total sum will be equal to 1.

Hence, the both issue are resolve of precious and probability . We not need to worry about on which scale the measurement was taken we will just put in it's range in which it layes like 5.2345455 and 5.2 both comes in 5 to 6 range for calculation we use integration Formula for discrete Random Variable:

$$\int P(xi) = 1$$

you there some thing funny if we want to get exact probability for a specific value it always be zero according to integration and we put range from $+\infty$ to $-\infty$ it will be one you can check it.

Now before we move to the function of probability distribution we should first probability distribution actually is.

So,

Definition 8 (Probability Distribution) *From the above we realise that each outcomes of any event or you can say it like each value of random variable have some probability attach to it. Now think it like if i plot all the random variables on the x-axis and probabilities on y-axis the plot is "Probability Distribution" of the random variable.*

It can be in form table also same outcome of random variable and probabilities as columns and each row represent a single outcome and its probability.

Now let's think so it like if you are interested in the chance of a specific event not all the possible outcome like in probability distribution .Like, you ask that what are the chance or probability that dice will give us a six , or what are the chance that dice give more then 4 it could be any thing a 4 , 5 or 6 no matter specifically any of them .

Here, comes the need of a probability Distribution Function.

There are three function:

1. Probability Mass Function (PMF).
2. Probability Density Function (PDF).
3. Cumulative Distribution Function(CDF).

Let's take thing simple from start we will just take about of the random variable which give fixed outcome means discrete distribution in which we know the chance for an exact output.

Definition 9 (Probability Mass Function (PMF)) *The function in which give the input of the outcome for which we want to know what are the chance that it will happen the probability of dice giving a 1 the 1 is given as the input the output is the probability of its occuring.*

But remember that the result will all whys be between 0 and 1 and some of all the outcomes must be equal to one

Formula :

$$p(X = x_i)$$

Most people cover first the PMF then PDf then at last the CDF the reason is that the CDF work for both but I will disccuss the CDF for discrete here and for continuous after the PDF.So let us move on to the next step.

Now what if neither what the full distribution not the specific for one outcome but instead for a range in discrete like if will be less then four This what the CDF solves

Definition 10 (Cumulative Distribution Function(CDF)) *it gives the probability for a combine value simple just add the chance of previous one in it like for 1 tells only for it when we move to 2 it will tell for both 1 and 2 which mean it can be either 1 or 2 . the graph will make it more clear.*

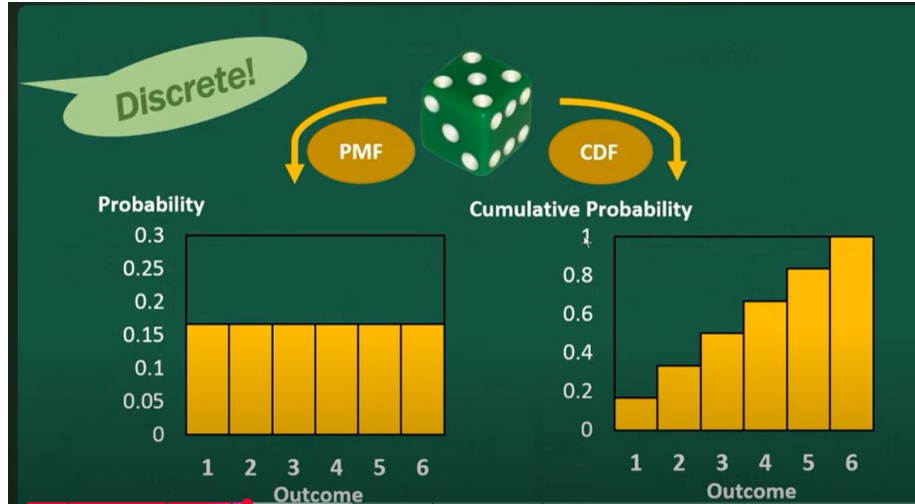


Figure 1: Difference between PMF and CDF for discrete variables.

Formula:

$$F(x) = P(X = x) = \sum_{x_i \leq x} P(X = x_i)$$

It will add the all probabilities upto that specific given number.

Now, its time to move for our other variable which are continuous variables in continuous variable we know that we can't give the exact probability to any variable the reason we discuss before it will be equal to the infinite if we assigned so take a range but we want the probability for a specific variable not a distribution.

Definition 11 (Probability Density Function (PDF)) *PDF is a function used for continuous variables that "tells the chance of values being near a point", not the exact value (since the chance of any exact value is zero). We use the "area under the curve" to find the chance that the value lies between two points.*

Formula:

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

Where:

$f(x)$ is the PDF

It must follow:

$$f(x) \geq 0$$

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

Definition 12 (Cumulative Distribution Function(CDF)) *CDF for continuous variables tells the total chance from the left up to a point x by adding (integrating) all chances from $-\infty$ to x .*

Formula:

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt$$

There are some concepts left before moving to the probability Distributions. So they are explain below:

Dispersion:- It tells the how is data spread. There are different types some of them are :

1. Variance(σ)
2. Standard Deviation
3. Inter Quartile Range (IQR)

Definition 13 (Variance) *Variance is the squared distance of data from the mean value. You can think of it like it tells about how the data is spread is there some uniformity in data or not.*

I am not explaining the other two you can see in the notice i explain there in detail

NOW, move to the next concept is very important and easy **Expected values**

Definition 14 (Expected Values) *It is the weighted average (arithmetic) mean. What it mean the probability of every outcome is also attached with it*

Formula for discrete Random variable :

$$E[x] = \sum_x x * P(X = x)$$

Formula for continuous Random variable :

$$E[x] = \int_x x.(f(x))$$

Now, the main point what is the point of doing this stuff means finding the Expected Value. The best and the most important which make the idea clear to me is the price bond given by the bank.

There are some price bond which are distributed to the people and after some month there is a lucky draw in which the person price bond which is selected get a price money and other get there money back . The think that make me curious was why does the bank do this they take loss by returning there money back and price two which is also of millions of dollar.

After learn about expected values i understand they not make loss instead they make profit . Let me explain it to you how this happen with expected value.

Let's suppose that the winning price of 1 million dollar will be given to 1 person. ten will be given 100 thousand dollar and 100 will be given the price of 10 thousand dollar. The price bond about 1 million are sold each worth 100 dollars each . the last thing is that the lucky draw will be done after 1 year and the loser will get there money back also. The amount the bank will get in total is \$100,000,000. The amount the bank will give as price money is :

$$\text{Price Money} = \$1,000,000 + (10 * \$100,000) + (100 * \$10,000) = \$3,000,000$$

Total Price to return to poeple is :

$$\text{Total Price} = \text{Original Price} + \text{Price Money}$$

$$\text{Total Price} = \$1,000,000,000 + \$3,000,000 = \$1,300,000,000.$$

As you know that the interest on each year is minimum 7% on orignal amount which is \$7,000,000 but bank give only \$3,000,000 so 4 million are there profit . But still the question remains that why poeple buy it and how does the expected value help.

Let's calculate the expected value for single person having 1 price bond worht \$100

1 in 1,000,000 chance to win \$1,000,000 = \$1

10 in 1,000,000 chance to win \$100,000 = \$1

100 in 1,000,000 chance to win \$10,000 = \$1

Expected Value which you will get is:

$$EV = \$1 + \$1 + \$1 = \$3$$

You pay \$100, but get back only \$3 on average.

You are only able to get \$3 on average the thing buys it that the 3% or \$3 are not give to every person this are give to just a few thatswhy it sound amazing and poeple think that they will be that choosen one poeple to win the price.

Thats the exact power of the expected value it tells you the relate that what you should expected from some time that sounds "too good to be true". This is the same which can be use by you to get the price that you can win in any gamble game this will save you from many thing that will make loose your monay or time.

The Next topic is the difference between liklihood and the probability they sound same but are not the same thing.

Probability vs Likelihood (With Intuition)

Conceptual Difference

Probability and Likelihood are two sides of the same coin, but they are used in opposite ways.

- **Probability** is used when we know the model or the conditions (called parameters), and we want to know the chance of getting a specific outcome.
- **Likelihood** is used when we already know the outcome (data we observed), and we are trying to figure out which condition or model (parameter) most likely produced this outcome.

Intuition: The Crime Scene Analogy

Imagine the following:

- **Probability:** You have a gun and you ask, “What kind of crimes could happen with this gun?” You are moving from the tool (parameter) to possible results (data).
- **Likelihood:** A crime has already happened. Now you’re asking, “What type of gun was used?” You are moving from the observed outcome back to the possible cause.

This analogy shows the direction of reasoning:

Probability: Cause (parameter) $\Rightarrow$ Effect (data) **Likelihood:** Effect (data) $\Rightarrow$ Cause (parameter)

Student-Friendly Explanation

- When you design a model (like tossing a biased coin), you assume certain parameters — e.g., a 70% chance of heads. If you ask “What’s the chance of getting 2 heads in 3 tosses?”, you are using **probability**.
- Now imagine you actually did 3 tosses and got 2 heads. You don’t know the real bias of the coin. You ask, “Which bias (60%, 70%, etc.) would most likely give me this result?” — this is **likelihood**.

In short:

- **Probability:** Predict outcomes based on known settings.
- **Likelihood:** Infer settings based on known outcomes.

Mathematical View

Let:

- X = observed data (e.g., 2 heads in 3 tosses)
- θ = parameter (e.g., probability of heads = 0.6)

Probability:

$P(X|\theta)$: What is the chance of seeing X if θ is true?

Likelihood:

$\mathcal{L}(\theta|X) = P(X|\theta)$: How likely is parameter θ given that we saw X ?

Even though the formula looks the same, the purpose is different: - In **probability**, θ is fixed, and X varies. - In **likelihood**, X is fixed, and θ varies.

Why Likelihood Matters in Machine Learning

In real-world data science or ML:

- You collect data (outcomes). - You build models with parameters (e.g., logistic regression with coefficients). - You want to find which parameter values make your model best match the data.

This is done using Likelihood — especially Maximum Likelihood Estimation (MLE):

$$\hat{\theta} = \arg \max_{\theta} \mathcal{L}(\theta|X)$$

That is, you try different θ values and choose the one that makes your observed data most likely to occur.

Log-Likelihood

Sometimes it is easier to work with the log of the likelihood, especially when multiplying many probabilities:

$$\log \mathcal{L}(\theta|X) = \log P(X|\theta)$$

This is called the log-likelihood and is often used in optimization.

Likelihood vs Maximum Likelihood Estimation (MLE)

1. Likelihood Intuition

Suppose we are testing a coin to check if it is fair. We flip it 10 times and observe the following:

Heads, Tails, Heads, Heads, Tails, Heads, Heads, Heads, Tails, Heads

We count:

- Heads = 7
- Tails = 3

Let p be the probability of getting heads. Our goal is to estimate p using the data we have.

Key Idea of Likelihood:

Likelihood tells us how likely a given parameter (like p) is to have produced the observed data. It flips the role of probability: instead of asking "What is the probability of this outcome given a model?", we ask "What is the best model (parameter) given the outcome?"

We try different values of p (e.g., 0.4, 0.5, 0.6, 0.7) and compute how likely each one is to produce the observed result (7 heads, 3 tails). The value of p that gives the **highest likelihood** is our best estimate.

2. Maximum Likelihood Estimation (MLE)

Definition:

Let $X = (x_1, x_2, \dots, x_n)$ be the observed data and θ be the unknown parameter.

Likelihood Function:

$$L(\theta | X) = P(X | \theta)$$

Maximum Likelihood Estimate:

$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} L(\theta | X)$$

This means: Find the value of θ that maximizes the likelihood function.

3. MLE for Coin Toss (Bernoulli Example)

Assume each toss is independent and follows a Bernoulli distribution with unknown p . The likelihood of observing 7 heads and 3 tails is:

$$L(p) = p^7 \cdot (1 - p)^3$$

This is the probability of getting 7 heads and 3 tails with probability p of heads.

To make calculation easier, we use the log-likelihood:

$$\log L(p) = 7 \log p + 3 \log(1 - p)$$

Take derivative and set it to 0:

$$\frac{d}{dp} [7 \log p + 3 \log(1 - p)] = 0$$

Solving this equation gives:

$$\hat{p}_{\text{MLE}} = \frac{7}{10} = 0.7$$

So, the most likely value of p (MLE estimate) is the proportion of heads in our data.

4. Summary of Intuition

- **Probability:** You know the model (parameter), and want to calculate the probability of outcomes.
- **Likelihood:** You know the outcomes (data), and want to find the best-fitting parameter.
- **MLE:** A method to choose the parameter that makes the observed data most likely.

In simple words:

MLE chooses the model that gives the highest score (likelihood) to the data you observed.